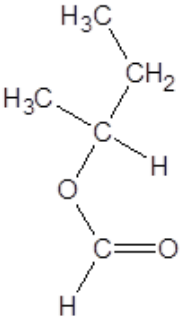


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		reflux (1) to avoid loss of reagents / products / solvent (1)	2			2		2
		(ii)		distillation (1) boiling temperature of ester is lower than alcohol/acid (1)	2			2		2
		(iii)		(conc) sulfuric acid	1			1		1
		(iv)		reagents $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3 + \text{HCOOH}$ (1) products  + H_2O (1)		2		2		

Question				Marking details	Marks available																																			
					AO1	AO2	AO3	Total	Maths	Prac																														
	(b)			award (1) each for any two of following • Na₂CO₃ / Mg - effervescence / fizzing / bubbles • acidified potassium dichromate - orange to green accept acidified potassium manganate(VII) - purple to colourless • iodine and sodium hydroxide / potassium iodide and sodium chlorate(I) - yellow precipitate award (1) for correct ✓/× for all three compounds for first test award (1) for correct ✓/× for all three compounds for second test award (1) for tests that enable all three compounds to be identified <table><tr><th>Reagent(s)</th><th>Observation expected for positive result</th><th>butan-2-ol</th><th>2-methylpropanoic acid</th><th>3-hydroxybutanoic acid</th></tr><tr><td>Na₂CO₃</td><td>effervescence / fizzing / bubbles</td><td>×</td><td>✓</td><td>✓</td></tr><tr><td>Mg</td><td>effervescence / fizzing / bubbles</td><td>×</td><td>✓</td><td>✓</td></tr><tr><td>acidified potassium dichromate</td><td>orange to green</td><td>✓</td><td>×</td><td>✓</td></tr><tr><td>acidified potassium manganate(VII)</td><td>purple to colourless</td><td>✓</td><td>×</td><td>✓</td></tr><tr><td>iodine and sodium hydroxide</td><td>yellow precipitate</td><td>✓</td><td>×</td><td>✓</td></tr></table>	Reagent(s)	Observation expected for positive result	butan-2-ol	2-methylpropanoic acid	3-hydroxybutanoic acid	Na ₂ CO ₃	effervescence / fizzing / bubbles	×	✓	✓	Mg	effervescence / fizzing / bubbles	×	✓	✓	acidified potassium dichromate	orange to green	✓	×	✓	acidified potassium manganate(VII)	purple to colourless	✓	×	✓	iodine and sodium hydroxide	yellow precipitate	✓	×	✓		5		5		5
Reagent(s)	Observation expected for positive result	butan-2-ol	2-methylpropanoic acid	3-hydroxybutanoic acid																																				
Na ₂ CO ₃	effervescence / fizzing / bubbles	×	✓	✓																																				
Mg	effervescence / fizzing / bubbles	×	✓	✓																																				
acidified potassium dichromate	orange to green	✓	×	✓																																				
acidified potassium manganate(VII)	purple to colourless	✓	×	✓																																				
iodine and sodium hydroxide	yellow precipitate	✓	×	✓																																				
				Question 9 total	5	7	0	12	0	10																														